

AYAN BANERJEE

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PROFESSIONAL SUMMARY

Computer Science graduate and MERN Stack Developer with hands-on internship experience building responsive React.js interfaces and RESTful Node.js/Express APIs in a live production startup environment, with hands-on exposure to Python and FastAPI. Skilled in JWT authentication, MongoDB data modeling, real-time communication with Socket.io, Docker containerization, CI/CD pipelines, and Git-based collaborative workflows. Strong problem-solving foundation in Java and JavaScript (ES6+), with a passion for designing scalable, secure full-stack applications and shipping features in agile sprint cycles. Currently learning and building with Large Language Models (LLMs), including AI API integration and prompt engineering, to create AI-powered features such as natural-language product recommendations.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Java, Python.

Frontend: HTML5, CSS3, React.js, Tailwind CSS, Responsive Design, Compatibility.

Backend: Node.js, Express.js, FastAPI, RESTful API Development, JWT Authentication/Authorization.

Real-Time & Architecture: Socket.io, WebSockets, Real-Time Communication, Modular/MVC Architecture, Centralized Error Handling.

Database: MongoDB, Mongoose, SQL, PostgreSQL, My SQL (coursework).

Frameworks: React.js, Express.js, Tailwind CSS, Next.js.

AI: Generative AI, LLM API Integration, Prompt Engineering.

Tools & DevOps: Git, GitHub, GitLab, Docker, CI/CD, GitHub Actions, VS Code, GCP, Vercel, Render, Agile/Sprint Workflows.

WORK EXPERIENCE

Web Development Intern — Vylex.ai

Dec 2025 – May 2026

Technology Startup | Remote

- Built and maintained responsive, cross-device frontend features using HTML5, CSS3, JavaScript, and React.js, contributing to a live production codebase.
- Collaborated with product and engineering teams in an agile environment, delivering UI components across weekly sprint cycles to meet release deadlines.
- Participated in peer code reviews and enforced Git best practices (feature branching, pull requests) to maintain code quality across the team.
- Debugged and resolved frontend issues reported during sprint testing, improving UI stability ahead of releases.

PROJECTS

SyncChat — Real-Time Chat Application | [Link](#)

React, Node.js, Express, Socket.io, MongoDB, Docker, CI/CD

- Implemented real-time messaging with Socket.io, including typing indicators, online/offline presence, and message delivery/read status.
- Developed REST APIs for message persistence, chat history, online-user tracking, and health monitoring using Node.js, Express, MongoDB, and Mongoose.
- Designed a modular backend architecture with controllers, services, routes, middleware, and socket handlers, along with centralized error handling.
- Containerized the application with Docker and configured CI/CD pipelines to automate build, testing, and deployment workflows.
- Deployed frontend and backend services and handled real-time connection, disconnection, and reconnection states for multi-client communication.

CarryCore — Full-Stack Bag E-Commerce Platform | [Link](#)

React.js, Node.js, Express.js, MongoDB, Tailwind CSS, JWT, LLM API Integration

- Architected a full-stack e-commerce application with responsive landing page, product catalogue, and cart UI using React.js and Tailwind CSS.
- Implemented JWT-based authentication and authorization with secure password hashing for user signup, login, and protected routes across frontend and backend.
- Designed RESTful APIs with Node.js and Express.js, with asynchronous request handling and error handling, using MongoDB for product catalogue and order management.
- Built reusable React components with client-side form validation and integrated state management for cart and session handling.
- Integrating an AI-powered shopping assistant using an LLM API to understand natural-language shopping requirements and recommend relevant products from the existing inventory.

Employee Management System (EMS) | [Link](#)

Node.js, Express.js, MongoDB, HTML5, CSS3, JavaScript

- Engineered a full-stack CRUD application for employee record management, featuring a responsive, user-friendly dashboard for efficiently listing, adding, and editing entries.
- Architected and implemented well-structured, RESTful backend APIs using Node.js and Express.js, leveraging MongoDB for reliable and persistent data storage.
- Enforced robust client-side form validation to strengthen data integrity, minimize invalid submissions, and improve overall application reliability.
- Deployed and maintained the application on Render, ensuring reliable public live-demo access for end users.

ACHIEVEMENTS

- Successfully completed the Google Cloud Arcade program, earning multiple Google Cloud Skill Badges and receiving official Google Cloud Arcade goodies for completing eligible challenge milestones.

EDUCATION

B.Tech - Computer Science and Engineering

Sep 2023 – July 2026

College of Engineering and Management, Kolaghat | MAKAUT University | CGPA: 7.1

Diploma - Computer Science and Technology

Mar 2020 – Jul 2023

Santiniketan Institute of Polytechnic | WBSCTE | CGPA: 8.32